

MANHATTAN, KS, USA

WMO: 724555

Lat: 39.135N

Lon: 96.679W

Elev: 322

StdP: 97.52

Time Zone: -6.00 (NAC)

Period: 96-19

WBAN: 03936

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-16.4	-13.3	-21.1	0.6	-14.7	-18.6	0.8	-12.3	10.7	9.3	9.3	6.9	1.8	10	0.522

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.5	37.7	24.2	36.0	24.2	33.9	23.8	26.0	33.7	25.4	32.9	24.8	31.9	5.2	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.0	19.6	29.8	23.0	18.5	28.9	22.6	18.0	28.5	82.3	33.8	79.4	32.7	76.6	31.9	28.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.7	9.1	8.1	-21.4	40.7	3.2	1.8	-23.6	42.0	-25.5	43.0	-27.3	44.0	-29.6	45.2		
			-21.7	27.7	3.1	0.7	-23.9	28.2	-25.7	28.6	-27.4	29.0	-29.7	29.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	13.0	-1.1	0.9	6.9	12.7	18.4	24.0	26.7	25.3	20.8	13.6	6.4	0.2
	DBStd	11.05	6.12	6.46	6.32	5.35	4.75	3.83	3.50	3.53	4.75	5.34	5.49	5.83
	HDD10.0	1239	346	261	135	32	2	0	0	0	0	25	133	307
	HDD18.3	2807	603	487	357	181	61	4	1	1	28	165	358	562
	CDD10.0	2319	1	6	38	113	263	420	518	473	323	136	26	2
	CDD18.3	846	0	0	2	12	63	174	260	216	101	17	1	0
	CDH23.3	9734	0	1	34	170	681	1977	3141	2399	1111	208	11	0
	CDH26.7	4617	0	0	5	42	246	906	1681	1207	472	58	1	0
Wind	WSAvg	3.3	3.1	3.4	4.0	4.2	3.6	3.6	3.0	2.6	2.9	3.1	3.2	3.0
Precipitation	PrecAvg	849	18	27	51	80	120	128	117	101	77	60	40	30
	PrecMax	1319	40	83	119	235	334	259	393	216	135	143	126	93
	PrecMin	619	2	2	0	24	28	9	47	19	11	5	1	1
	PrecStd	161	10	20	31	43	80	58	77	42	33	33	35	26
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	18.0	22.0	26.9	30.9	34.0	37.7	40.2	40.2	37.2	31.4	25.0	18.8
		MCWB	9.7	12.5	16.1	19.2	21.9	24.1	24.1	23.5	22.6	21.0	15.2	14.2
	2%	DB	14.7	17.7	23.7	27.6	31.4	35.3	38.0	37.4	33.7	27.9	22.0	15.1
		MCWB	8.1	10.6	14.7	17.3	21.3	24.1	24.5	24.0	22.9	18.2	14.4	9.1
	5%	DB	11.6	14.2	21.0	25.2	29.4	33.5	36.3	34.9	31.7	25.6	18.8	12.4
		MCWB	6.4	9.3	13.3	16.3	20.3	23.8	24.6	24.0	22.4	17.5	12.6	7.9
	10%	DB	8.6	11.5	17.9	22.6	27.3	31.8	34.1	32.7	29.2	22.9	16.3	9.5
		MCWB	4.6	6.8	11.8	15.2	19.8	23.3	24.3	23.6	21.1	16.5	11.2	5.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.7	14.5	18.5	20.2	24.1	26.5	26.9	26.7	25.1	22.6	17.2	14.8
		MCDB	14.8	18.7	24.7	26.9	30.7	34.1	34.6	33.4	32.4	27.9	21.7	17.7
	2%	WB	8.6	11.9	16.6	18.9	22.8	25.5	26.2	25.7	24.1	20.8	15.4	10.8
		MCDB	13.5	16.1	20.7	24.9	29.2	32.9	34.2	32.8	31.1	25.9	19.2	13.1
	5%	WB	6.7	9.3	14.7	17.6	21.7	24.7	25.6	25.0	23.2	19.1	13.7	7.9
		MCDB	11.3	14.7	19.6	23.7	27.7	31.7	33.5	32.1	29.9	23.6	18.4	11.7
	10%	WB	4.7	6.8	12.2	16.0	20.6	23.8	25.0	24.2	22.2	17.3	11.5	5.8
		MCDB	8.8	11.1	18.0	21.8	25.9	30.2	32.4	31.0	28.0	21.6	15.8	9.4
Mean Daily Temperature Range	5% DB	MDBR	12.5	12.4	13.5	13.5	12.7	12.4	12.5	12.7	13.5	13.8	13.3	12.1
		MCDBR	18.0	18.0	18.6	16.8	15.2	13.9	14.8	15.5	14.7	16.7	17.7	16.8
		MCWBR	12.1	11.7	10.6	9.0	7.0	5.6	4.7	5.1	5.9	8.6	10.6	11.5
	5% WB	MCDBR	17.0	16.6	14.4	14.4	13.1	12.6	12.8	13.1	12.9	13.1	14.6	15.1
		MCWBR	11.8	11.4	8.8	8.7	6.8	5.9	5.0	5.5	5.9	8.0	9.9	11.0
Clear-Sky Solar Irradiance	taub	0.287	0.302	0.333	0.379	0.410	0.421	0.428	0.427	0.383	0.342	0.301	0.282	
	taud	2.483	2.449	2.409	2.304	2.288	2.291	2.296	2.298	2.374	2.464	2.513	2.501	
	Ebn at Noon	895	925	925	896	872	859	849	841	860	866	870	874	
	Edh at Noon	78	92	107	127	131	131	130	126	110	90	75	71	
All-Sky Solar Radiation	RadAvg	2.32	3.09	4.05	5.14	5.87	6.66	6.56	5.86	4.84	3.54	2.55	1.99	
	RadStd	0.20	0.31	0.30	0.32	0.47	0.39	0.45	0.29	0.31	0.37	0.21	0.19	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (27 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page